



Interpreting pie charts

Task A: Calculating frequencies from the size of a sector

subject	frequency	angles
maths		152°
English		60°
science		100°
other		48°
total		360°



1) A group of students were asked what their favourite subject was.

25 students voted for "science".

a) Calculate the **multiplier** for this set of data.

b) Calculate all other frequencies by dividing using this **multiplier**.

c) Calculate the total number of students who were asked.

2) Oakfield Academy are organising a fair to raise money to improve environmental efficiency of their school.

The data shows votes of students for what they will do during the fair.

Calculate all missing frequencies.

activity	frequency	angles
cake sale		63°
concert	39	117°
charity race		150°
face painting		30°
total		360°

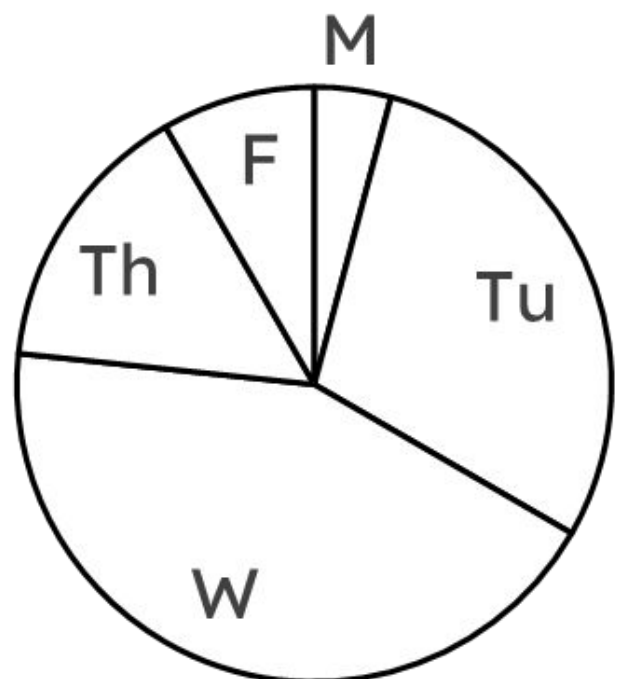


day	freq.	angles
Monday	5	
Tuesday		
Wednesday		156°
Thursday		54°
Friday		

3) The pie chart shows the suggested days of the week the fair could be run on.

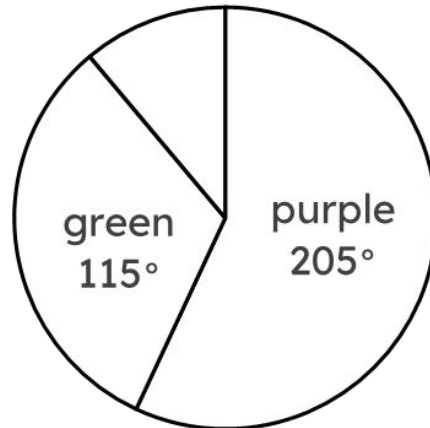
Complete the table by:

- measuring all missing angles,
- calculating all missing frequencies.

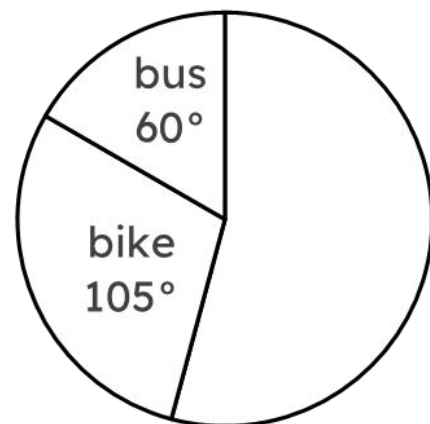


4) Fill in all missing information for each table.

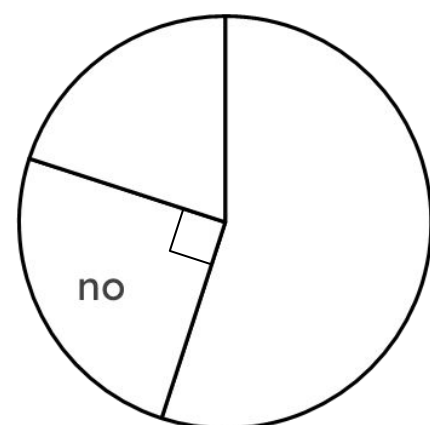
colour	freq.	angles
purple		
green		
blue	8	
total		



vehicle	freq.	angles
car	13	
bike		
bus		
total		

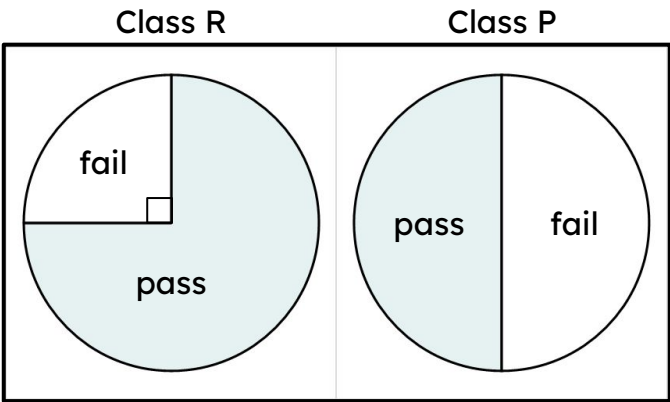


vote	freq.	angles
yes	165	
no	75	
undecided		
total		



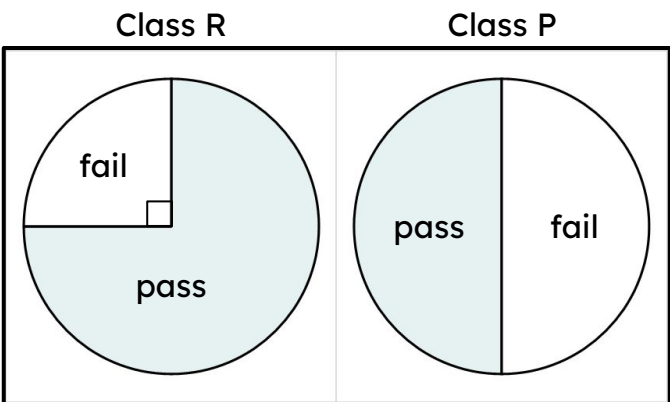
Task B: Comparing pie charts

1) A headteacher is comparing the exam results of two classes.



The headteacher says “more students passed the exam in Class R”.

The headteacher may be incorrect. Can you explain why?



2) In both classes, 15 students passed the exam.

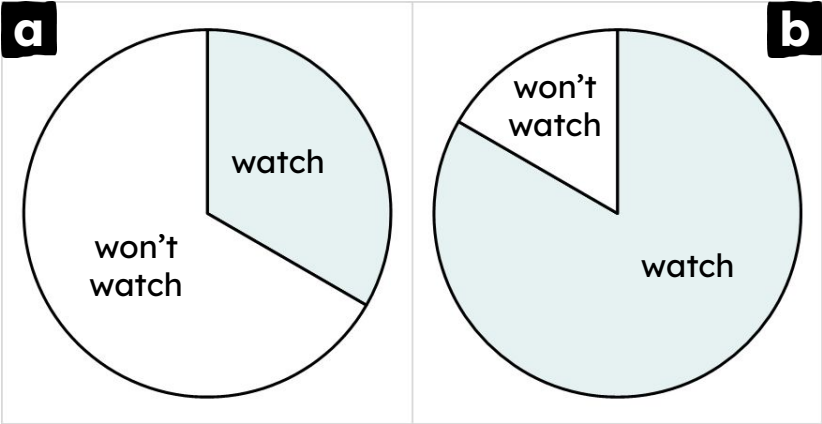
Complete the ratio tables below to work out which class had more students.

Class R		
pass	fail	total
		360°
15		

Class P		
pass	fail	total
180°	180°	360°
15		

3) People in Leeds city centre were asked whether they support Leeds United, and whether they would watch Leeds United’s next football match.

Two pie charts were drawn, one to show supporters, and one to show non-supporters.

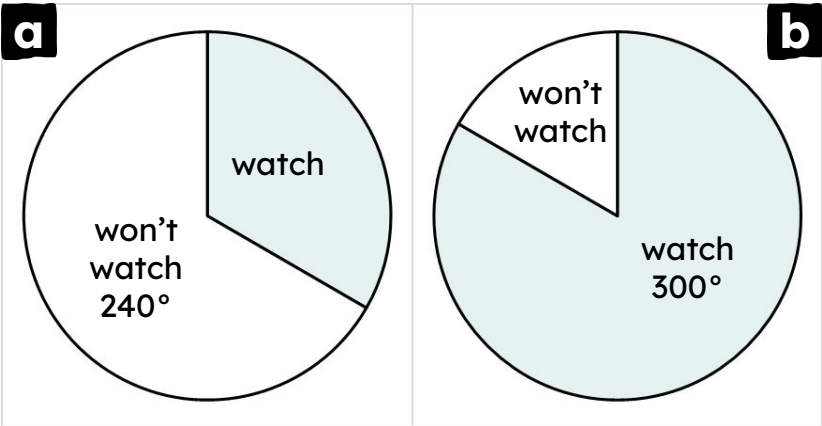


Which of the two pie charts represents the group of people who support Leeds United? Explain how you know.

4) The minor sector of each group both represent 80 people.

In which group were more people interviewed, and by how many people?

Fill in the ratio tables to help.



group a

watch	won't watch	total
		360°
80	160	

group b

watch	won't watch	total
		360°
	80	